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Project Title:	Freely Suspended Liquid Flims and their Applications in Biological Research		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
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Space Biology Element:	None		
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Comments:			
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Flight Program:			
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Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		
Task Description:	Liquid films made of surfactants and phospholipids will be used as a nearly two-dimensional (2D) suspension medium. Mechanical properties of these films, such as viscosity, surface tension and elasticity will be investigated. We will also incorporate biological macromolecules, such as proteins and DNAs, into these films and study their structures and dynamics in films that have thickness comparable to the molecular size. In high concentrations, biomolecules will order spontaneously forming 2D crystals that are only one molecular layer thick. We will concentrate on using weak shear flows and electric fields to influence 2D crystallization so that large, high-quality crystals can be produced for structural studies. The molecular inclusions can also be used as anchoring points for cells growth. The main research tools used in the experiments are molecular biology, light scattering and imaging techniques. The proposed research is aimed at better understanding of physical and biological properties of thin fluid membranes in the presence of macromolecular inclusions, such as colloidal particles, proteins and DNAs. The enhanced understanding		

	will facilitate new technological development in areas such as membrane rheology, micro-scale particle sorting, and two-dimensional protein crystallization. The technique can also be beneficial for patterning surfaces on microscopic scales, which has find increasing applications in areas such as cell growth, drug screening, and biochip technology. Develop new techniques by which biological molecules can be characterized and manipulated in ultra-small quantities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This research has covered broad areas of research in biotechnology and biophysics. We started out by investigating the effect of acoustic vibrations on the growth and ordering of two-dimensional colloidal particles. The rational was that if we could order these small uniform sized particles (<1.0 mm in diameters) in a thin layer of fluid, one could build a large 2D single-domain crystal. The technique is therefore potentially useful to crystallize 2D proteins including membrane proteins, which until recently had no systematic method of crystallizing them. We have demonstrated that indeed small colloidal particles can be forced to crystallize on a large scale (1cmx1cm) by applying appropriate amounts of vibration energy and frequency. In essence one could use the acoustic waves to align domains of poly-crystals until a single crystal is formed. This work was published in Applied Physics Letter. In a subsequent work we also studied the micro-defect structures of the 2D crystals as functions of vibration energy and time. It was found that the defect structures could be characterized by small-angle and large-angle grain boundaries and were in quantitative good agreements with the theoretical predictions made by Shockley and Read about 50 years ago. The length of the grain boundaries was found to diminish logarithmically in time for small vibration amplitudes but for large vibration amplitudes, a steady state could be reached where the total length was constant at long times. This work is currently under review. The study of thin liquid films also led us to the investigation of chemical modifications to solid substrates that are suitable for cell culture. We have explored the possibilities of patterning neurons using a thin layer of polylysine on glass cover slips. Previous experiments have demonstrated that neuronal cells are capable of growing on thin layer of polylysine but are inhibited on agarose. This difference allows us to pattern neurons and to build small neuronal circuits. Our first experiment was carried out by simply drawing patterns on glass substrates using a small capillary tube a few microns in diameters, which was filled with polylysine. This so called micro-fountain-pen technique showed some promise but it was very slow and the successful rate was also low. To make the technique more practical such that it can be reliably used in the laboratories, a new improved technique is needed. The technique we are currently using combines micro-lithography and biochemical surface modifications. Briefly, one first spin-coats on a small glass disk (a few centimeters in diameter) with polylysine, which are short peptides that bind strongly to the glass substrate and to biological cells including neurons. One then creates barriers that inhibit cell growth. The physical properties of agarose are particularly convenient to work with because it is a fluid at high temperatures but forms a solid gel at low temperatures. We used the micro-lithographic techniques to create small stamps with micron sized features to protect areas of polylysine where cells are allowed to grow and let the rest of areas on the glass flooded with agarose and let to solidify. We were able to produce patterns with feature size of 300 microns and are striving to reduce this size down to 100 microns, which would allow isolation of individual neurons. Preliminary results are very promising in that using 3T3 cells, which are more convenient to grow than neuronal cells in the lab. We observe that the cells attach strongly in the wells surrounded by the agarose and they proliferate as in normal cell cultures. Everywhere else on the glass plate, there is no 3T3 cells, demonstrating the concept of the technique. Our next step is to work with primary neural cell cultures and to build small networks of neurons, to be discussed below. In collaboration with Prof. G.Q. Bi of the University of Pittsburgh Medical School (UPMC), we also commenced a study of vesicle dynamics in synapses of central nerve systems. This work is made possible by the recruitment of Mr. Matt Shtrahman who has a strong background in biology and also has keen interest in using modern physical techniques to address important problems in neurobiology. Why vesicle dynamics in synapses are interesting to study? It is well known that neurons are connected by synapses, which serves as a chemical relay station converting an electric signal (an action potential) from the pre-synaptic neuron into chemical signal by releasing neuron transmitters. In order to release neuron transmitters, the vesicles, which encapsulate the neuron transmitters, needs to be transported to active zone adjacent to the postsynaptic neuron. This process of transport is not well understood partly due the lack of techniques to investigate the phenomenon. We note that the synapses in central nerve system are very small, about 1 mm in diameter and 0.5 mm in thickness. The vesicles are even smaller with diameters (40 nm) well below the diffraction limit of optical microscope. Thus direct observation of vesicle motion is out of question. The physical process of vesicle movement however is an important one because the fidelity of neuron transmission depends on their mobility and has strong implications for learning and memory. To this end we have implemented a fluorescence correlation spectroscopic technique to study motion of vesicles inside synapses. A small fraction of vesicles in individual synapses are labeled by FM1-43 dye via endocytosis, the neurons are then destained and are ready for observation. A tiny laser spot impinges on the synapses and fluorescence intensity fluctuations due to motion of dye-label vesicles is measured by a digital correlator. We found that vesicle motion is highly restricted in the sense that if one assigns an effective viscosity to the cytosol, it would be 1000 times greater than water. However several lines of evidence showed that the slow motion of vesicles is not due to the high viscous environment but rather to the stickiness of the vesicles to cytoskeleton elements. Using okidaic acid, which dissociates vesicles from their corresponding binding sites, the vesicle motion increases by nearly 2 orders of magnitude, indicating that the viscosity of the cytosol in the synapses is not very different from that of normal cellular environments such as inside the soma of neuronal cells. This work is submitted to the rapid communication of Neuron Science and is currently under review. In the near future the fluorescence correlation spectroscopy (FCS) technique used in this experiment will be combined with the neuron patterning techniques discussed above to study network properties of a small cluster of neurons. This would open up many new research opportunities in neuron science and in drug discovery.
Bibliography Type:	Description: (Last Updated: 05/08/2019)
Articles in Peer-reviewed Journals	Shtrahman M, Yeung C, Nauen DW, Bi GQ, Wu XL. "Probing vesicle dynamics in single hippocampal synapses." Biophys J. 2005 Nov;89(5):3615-27. Epub 2005 Aug 19. Erratum in: Biophys J. 2006 Jan 15;90(2):708. https://doi.org/10.1529/biophysj.105.059295 ; PubMed PMID: 16113110 ; PubMed Central PMCID: PMC1366854 , Nov-2005

Articles in Peer-reviewed Journals	Wei QH, Wu XL. "Grain boundary dynamics under mechanical annealing in two-dimensional colloids." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Aug;70(2 Pt 1):020401. Epub 2004 Aug 31. https://doi.org/10.1103/PhysRevE.70.020401 ; PubMed PMID: 15447470 , Aug-2004